Knotted Topoi: the lift of the knotted set-theoretic universe, and fully abstract denotational semantics for the category of graph-structured lambda theories

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Abstract

We lift the knotted set-theoretic universe of *A Knotted Universe* — two mutually recursive copies of set theory tied so that each colour's atoms are the other colour's sets — one categorical level, to a *knotted topos*: four sort-topoi (red sets and atoms, black sets and atoms) tied by two geometric knots and an involutive colour-swap, with the reflection law $*@P \equiv P$ become the involutivity of an equivalence of topoi. This is the principal result. As an application we transport the operational view of computation to a denotational one. The MeTTaIL language presents the category of finitely presentable graph-structured lambda theories (GSLTs); desugaring MeTTaIL into the rho calculus, and composing with the fully abstract rho semantics of *Quoting is Colour-Swap* lifted into the knotted topos, gives a compositional denotation $[\![-]\!]_{\mathcal{K}}$ that is fully abstract for context bisimulation: $[\![P]\!]_{\mathcal{K}} = [\![Q]\!]_{\mathcal{K}} \iff P \sim Q$. Because GSLTs subsume every known classical model of computation — the λ-calculus and hence PCF, CCS, the π-calculus [16], the ambient calculus, SKI, Turing machines, and term rewriting, with universal algebra as the dynamics-free slice — the construction supplies a fully abstract denotational semantics for all of them at once, at least in the finitely presentable case. When Milner found the full abstraction problem for PCF he is said to have quipped, "so much for denotational semantics"; the linear-logic and game-semantics programme answered him for PCF, and the knotted topos answers him uniformly: the model is the bisimulation quotient itself — the final coalgebra of a behaviour functor internal to the topos — so it is fully abstract by construction, compositionality being the theorem and the context-labelled (Leifer–Milner) presentation being what makes bisimulation a congruence. The keystone is that the OSLF-generated Hennessy–Milner logic is the internal logic of the knotted topos: adequacy is the statement that the subobject classifier classifies bisimilarity. The risk ledger of the source papers transfers: finitely presentable theories cost no infinite risk, recursion costs one ω — black infinity — and not one primitive name.

Contents

1 Introduction

1.1 Milner's lament

In the nineteen-seventies Milner and Plotkin gave PCF — a typed λ-calculus with arithmetic and recursion — both an operational and a denotational semantics, the latter in Scott's domains, and asked whether the two agree exactly: whether two terms receive the same denotation precisely when no program context can tell them apart. They do not. The Scott model is not *fully abstract*; the celebrated witness is "parallel or", a function definable in the model but not in the language, so the model distinguishes terms the language equates [7, 8, 9]. The *full abstraction problem* for PCF — to build a model that is neither too coarse nor too fine — proved deep and durable, and Milner is said to have summarised the discovery with a quip: "so much for denotational semantics."

The linear-logic and game-semantics programme answered him for PCF two decades later. Reading a type as a game and a program as a strategy, and cutting the model down to the strategies that are *innocent* and *history-free*, Abramsky–Jagadeesan–Malacaria and Hyland–Ong built genuinely fully abstract models of PCF [10, 11], in the spirit of Girard's linear logic [12]: definability is recovered exactly, the parallel-or is excluded by the discipline on plays, and observational equivalence is, on the nose, equality of strategies modulo the game's own equivalence. The lesson of that programme is that the right denotational object is an *interaction* structure — plays between a player and an opponent — rather than a space of values.

1.2 The broader result

This paper makes the game-semantic moral uniform, and in doing so answers Milner not for one language but for a category that contains them all. A *graph-structured lambda theory*

(GSLT) [4] is a triple (grammar, equations, rewrites): freely generated terms as states, equations identifying states, rewrites animating them. The class of GSLTs subsumes every known classical model of computation. The λ-calculus is the GSLT with β as its base rewrite, and PCF is that GSLT enriched with constants, conditionals and a fixpoint; CCS, the π-calculus, the ambient calculus and the rho calculus are GSLTs whose base rewrite is communication; SKI is a GSLT of combinator contraction; a Turing machine is a GSLT of tape rewrites; and a term-rewriting system is a GSLT outright [30]. Universal algebra and the Lawvere theories sit inside as the dynamics-free slice, where the rewrite component is empty [4].

For all of these — at least the finitely presentable ones — a single construction supplies a fully abstract denotational semantics. The construction does not search for a model that happens to be neither too coarse nor too fine; it takes the model to *be* the bisimulation quotient, the final coalgebra of a behaviour functor, and proves that this quotient is a compositional, internally topos-theoretic object. Full abstraction is then by construction: the failure that defeated the Scott model — a model element with no program realising it — cannot occur, because every element of the final coalgebra is the behaviour of a process, and two processes are identified exactly when bisimilar. What had to be *proved* is the other half, compositionality: that the denotation is a homomorphism and that bisimulation is a congruence. The second source paper supplies precisely this, by labelling transitions with contexts after Leifer and Milner [13, 14], so that the bisimulation game already ranges over every environment and congruence holds with no closure step [2]. The interaction structure the game semanticists found necessary is, here, built in from the ground: the knot reads red as player and black as opponent [1], and the denotation lives in their interaction.

1.3 The two parts

This is the third in a sequence. *A Knotted Universe* [1] builds a universe of reflective set theories from $\varnothing$ alone, as a red copy $\mathsf{Set}[X]$ and a black copy $\overline{\mathsf{Set}}[\overline{X}]$ tied by $X = \overline{\mathsf{Set}}[\overline{X}]$, $\overline{X} = \mathsf{Set}[X]$, each colour's atoms the other's sets. *Quoting is Colour-Swap* [2] gives a fully abstract denotation of the rho calculus [5] there, refining the knot into four sorts with quoting realised as a colour-swap and proving full abstraction for the resulting final-coalgebra semantics. The present paper has a principal result and an application.

Principal result (§3): the lift to knotted topoi. We categorify the four-sorted model. The four sorts become four topoi $\mathcal{S}, \mathcal{A}, \overline{\mathcal{S}}, \overline{\mathcal{A}}$; the two knots become geometric equivalences; the colour-swap becomes an involutive auto-equivalence with $\mathsf{s} \circ \mathsf{s} \cong \mathrm{id}$; quoting and dereference are derived, $@ = \kappa \circ \mathsf{s}$, $* = \mathsf{s} \circ \kappa^{-1}$, and $*@P \equiv P$ becomes the involutivity of an equivalence of topoi. The knotted topos $\mathcal{K}$ is the (bi)fixpoint of the two-sorted endofunctor of [1, §8.1] read in the 2-category $\mathfrak{Top}$ of Grothendieck topoi. Its finitary core is set-sized, its full version class-sized, exactly along the ledger line of the source papers. The behaviour functor lifts to an internal endofunctor whose final coalgebra is the process object inside $\mathcal{K}$, where colour-blind non-well-foundedness supplies recursion and finality makes bisimilar processes internally equal. The OSLF-generated logic is the internal logic of $\mathcal{K}$.

Application (§4): operational to denotational. MeTTaIL (rholang 1.4) [6] — a language of types, equations-on-types and rewrites-on-types — presents the category $\mathbf{GSLT}^{\mathrm{fp}}_{\sim}$ of finitely presentable GSLTs: a MeTTaIL program *is* a finite GSLT, and a MeTTaIL language map is a GSLT morphism. Desugaring MeTTaIL into core rho (§4.2, Appendix A) installs each base rewrite $L \Rightarrow R$ as a guarded receiver at the channel naming the redex's location,

$$[\![L \Rightarrow R]\!](c) \;=\; \mathsf{for}([\![L]\!] \Leftarrow c)\{\, c\,!\,([\![R]\!])\, \},$$

and composing with the rho denotation lifted into $\mathcal{K}$ yields a compositional $[\![-]\!]_{\mathcal{K}}\colon \mathbf{GSLT}^{\mathrm{fp}}_{\sim} \to \mathcal{K}$ that is fully abstract for context bisimulation. This is the transport of the operational view to the denotational view, and the fully abstract model promised above.

1.4 Two standing conventions

MeTTaIL is used as the legible surface for writing rules and as the presentation of $\mathbf{GSLT}^{\mathrm{fp}}_{\sim}$; it is *eliminable sugar* over core rho, with its equation component factored through Church encodings, and the topos-theoretic recapitulation is carried out in core rho after the sugar is compiled away. Nothing in the four-topos picture depends on a MeTTaIL primitive. As to optimality: the channel c above is the sound, non-optimal reflection of the location, not the optimal set-automaton state of [3]; optimality is in tension with what a channel must do here (keep distinct runtime locations distinct), is unneeded for the lift, and is deferred together with the channel-naming correction noted in [3]. The optimal and the present scheme induce the *same* context-labelled transition system, so the choice is invisible to everything downstream.

We write in the programmatic-with-obligations style of the source papers [1, 2]: definitions and constructions are pinned down, the load-bearing existence, coherence and calibration facts are collected as numbered obligations (§5), and for the finitary core each reduces to a statement about hereditarily finite data.

2 Recap

We fix notation; [1, 2, 4] carry the details.

2.1 The knotted universe and its four sorts

The red and black copies of set theory are tied by $X = \overline{\mathsf{Set}}[\overline{X}]$, $\overline{X} = \mathsf{Set}[X]$. [1, §8.1] exhibits the tie as the fixpoint of the two-sorted endofunctor on a category of classes

$$\mathsf{H}(R, B) \;=\; \big(B + \mathcal{P}R,\;\; R + \mathcal{P}B\big), \tag{1}$$

whose initial algebra $(X, \overline{X})$ satisfies $X \cong \overline{X} + \mathcal{P}X$, $\overline{X} \cong X + \mathcal{P}\overline{X}$. The initial algebra is the well-founded reading (ZFA/FM per colour); the final coalgebra is the non-well-founded reading, in which colour-blind membership $\in_\star$ admits alternating-colour infinite descent [1, §7]. The model M of [1, Thm. 1] is a nominal set [27, 26], hence lives in the Schanuel topos [1, §6.1].

[2] keeps four sorts distinct: red sets S, red atoms A, black sets $\overline{S}$, black atoms $\overline{A}$, with knots $\kappa\colon \overline{S} \xrightarrow{\cong} A$, $\overline{\kappa}\colon S \xrightarrow{\cong} \overline{A}$, and a polymorphic involution s with $\mathsf{s} \circ \mathsf{s} = \mathrm{id}$, homomorphic for the set operations and coherent, $\mathsf{s} \circ \kappa = \overline{\kappa} \circ \mathsf{s}$. With processes red sets and names red atoms,

$$@ := \kappa \circ \mathsf{s}\colon S \to A, \qquad * := \mathsf{s} \circ \kappa^{-1}\colon A \to S, \tag{2}$$

and $*@P \equiv P$ is the calculation $\mathsf{s} \circ \kappa^{-1} \circ \kappa \circ \mathsf{s} = \mathsf{s} \circ \mathsf{s} = \mathrm{id}$ [2, Prop. 6].

2.2 The rho model

The rho grammar is $P ::= 0 \mid P \mid Q \mid \mathsf{for}(y \Leftarrow x)P \mid x!(Q) \mid *x$, $x ::= @P$; there are no primitive names and no restriction ν, and recursion enters through reflection rather than replication [2, §2]. Transitions are labelled by contexts $P \xrightarrow{F} P'$, F a minimal environment (an idem-pushout after Leifer–Milner) in which P completes a reaction. The behaviour functor is $\mathsf{B}(R) = \mathcal{P}(\partial T_K \times R)$, ∂T_K the Fire context-labels; the process universe is $\mathsf{Proc} = \nu\mathsf{B}$, and bisimilar processes are equal in it. Full abstraction $[\![P]\!] = [\![Q]\!] \iff P \sim Q$ holds for a congruence, read through OSLF adequacy as agreement of ∂T_K-indexed Hennessy–Milner formulae [2, §6].

2.3 GSLTs, the K-family, and the three views of ∂T

A GSLT is (grammar, equations, rewrites) [4]. An *interactive* GSLT carries a distinguished family K of constructors — the heads of the hypothesis-free (*base*) rewrites, which are the interactions; rules with hypotheses are propagation discipline. A *location* is a splitting (k, t) of a term into a context k and a subterm t, with the cut between them as the place; the splittings $\partial T \times T$ fibre over T by plugging, and the sub-bundle *Fire* consists of those whose subterm matches a base LHS. The one object ∂T_K is read three ways: as Firing labels, as Witnessing indices of OSLF modalities $\langle F \rangle \varphi$, and as Placing of interactions in term structure. The two categories of GSLTs differ in what morphisms preserve: the *syntactic* category preserves term structure, the *semantic* category $\mathbf{GSLT}_\sim$ preserves bisimulation. We work throughout with the semantic category, and write $\mathbf{GSLT}^{\mathrm{fp}}_\sim$ for its full subcategory of finitely presentable objects.

3 Principal result: the knotted topoi

Throughout $\mathfrak{Top}$ is the 2-category of Grothendieck topoi, geometric morphisms, and geometric transformations; $f^* \dashv f_*$ is a geometric morphism and $\simeq$ equivalence of topoi [21, 22].

3.1 Four topoi, two knots, one swap

Definition 1 (The sort topoi)**.** The four sorts lift to four topoi — $\mathcal{S}$ (red sets), $\mathcal{A}$ (red atoms), $\overline{\mathcal{S}}$ (black sets), $\overline{\mathcal{A}}$ (black atoms) — organised as a fibration $p\colon \mathcal{K} \to \mathcal{G}$ over the four-object index category $\mathcal{G}$ whose objects are the sorts and whose non-identity arrows are the kind-coercions $\mathcal{A} \to \mathcal{S}$, $\overline{\mathcal{A}} \to \overline{\mathcal{S}}$ together with the colour edges. The fibre over a sort is that sort's topos; the total category $\mathcal{K}$ is the *knotted topos*, and p is its *grading*.

Definition 2 (Knots as geometric coercions)**.** The two knots lift to geometric equivalences

$$\kappa\colon \overline{\mathcal{S}} \xrightarrow{\simeq} \mathcal{A}, \qquad \overline{\kappa}\colon \mathcal{S} \xrightarrow{\simeq} \overline{\mathcal{A}},$$

presenting a black-sets topos as the red-atoms topos and vice versa. The presentation is *opaque*: the geometric inclusion of an atom topos into the ambient sheaf topos forgets the other colour's interior, exactly the forgetfulness by which the atoms of [1, §8.1] "acquire no members". Opacity is colour-opacity [1, Rem. 2].

Definition 3 (The swap as an involutive auto-equivalence)**.** The colour symmetry lifts to a polymorphic, sort-respecting equivalence

$$\mathsf{s}\colon \mathcal{S} \simeq \overline{\mathcal{S}}, \qquad \mathsf{s}\colon \mathcal{A} \simeq \overline{\mathcal{A}}, \qquad \mathsf{s} \circ \mathsf{s} \cong \mathrm{id},$$

homomorphic for the geometric structure (initial object, coproducts, subobject classifier), with the coherence 2-cell

$$\mathsf{s} \circ \kappa \;\cong\; \overline{\kappa} \circ \mathsf{s} \qquad (\text{both } \overline{\mathcal{S}} \to \overline{\mathcal{A}}). \tag{3}$$

Proposition 4 (Reflection, lifted; sketch)**.** *Defining* $@ := \kappa \circ \mathsf{s}\colon \mathcal{S} \to \mathcal{A}$ *and* $* := \mathsf{s} \circ \kappa^{-1}\colon \mathcal{A} \to \mathcal{S}$ *as geometric morphisms, one has* $* \circ @ \cong \mathsf{s} \circ \mathsf{s} \cong \mathrm{id}$ *and* $@ \circ * \cong \mathrm{id}$*, so the model validates* $*@P \equiv P$ *and* $@{*}x \equiv_N x$ *not by an imposed equation but as the involutivity of an equivalence of topoi.*

Proof sketch. The middle isomorphisms are the unit/counit of the knot equivalence (Definition 2) and the involutivity 2-cell (Definition 3); coherence is (3). This is [2, Prop. 6] one level up, equality of maps replaced by canonical isomorphism; the triangle and pentagon coherences are Obligation 2. □

3.2 The knotted topos as a bifixpoint

Construction 5 (The knot 2-functor)**.** Let $\mathcal{P}\colon \mathfrak{Top} \to \mathfrak{Top}$ be the 2-functor sending a topos to its topos of internal sheaves on the canonical (power) site — the topos-level analogue of the powerclass operator of (1) [19, 21] — and let $\ltimes$ denote the atom-coproduct adjoining one topos to another as a topos of opaque urelements (the geometric analogue of $A + (-)$, Definition 2). Define the endo-2-functor on $\mathfrak{Top} \times \mathfrak{Top}$

$$\widehat{\mathsf{H}}(\mathcal{R}, \mathcal{B}) \;=\; \big(\mathcal{B} \ltimes \mathcal{P}\mathcal{R},\;\; \mathcal{R} \ltimes \mathcal{P}\mathcal{B}\big). \tag{4}$$

A *red/black pair of knotted topoi* is a (pseudo-)fixpoint $(\mathcal{E}, \overline{\mathcal{E}})$ of $\widehat{\mathsf{H}}$:

$$\mathcal{E} \;\simeq\; \overline{\mathcal{E}} \ltimes \mathcal{P}\mathcal{E}, \qquad \overline{\mathcal{E}} \;\simeq\; \mathcal{E} \ltimes \mathcal{P}\overline{\mathcal{E}},$$

each colour's topos its own sheaves with the other's as opaque atoms. The four-sort fibration of Definition 1 is read off this pair: $\mathcal{S}$ and $\mathcal{A} = \kappa\overline{\mathcal{S}}$ from $\mathcal{E}$, and $\overline{\mathcal{S}}$, $\overline{\mathcal{A}} = \overline{\kappa}\mathcal{S}$ from $\overline{\mathcal{E}}$.

Remark 6 (Which fixpoint: the knob is the source papers'). The *initial* algebra of $\widehat{\mathsf{H}}$ is the well-founded reading; each colour's topos models ZFA/FM internally, the picture of [1, §5] lifted. The *final* coalgebra is the non-well-founded reading, admitting the alternating-colour descent of [1, §7], and is the reading the process semantics needs (§3.3). For the *finitary* core — $\mathcal{P}$ replaced by its finite-power analogue — the fixpoint exists at the level of topoi as the colimit/limit of the evident ω-chain, set-sized; for the full $\mathcal{P}$ it lives in the class-sized framework, along the finitary/infinitary ledger line of [1, §5.5–5.6, §8.1]. Existence is Obligation 1.

Remark 7 (The grading is the modality, internalised). [1, §6.5] and [2, Rem. 8] ask for a graded modality separating colour-respecting (equivariant) from reflective computation. The fibration $p\colon \mathcal{K} \to \mathcal{G}$ *is* that grading: a computation over a single colour $\{\mathcal{S}, \mathcal{A}\}$ using only $@, *$ never observes black structure — the equivariant fragment, ordinary nominal names with every guarantee intact — while applying s to a name's interior crosses a colour edge, and reindexing along that edge records the reflective surplus. Decidable structural identity of names is the decidability of equality in $\mathcal{A}$, settled out of red's sight by black equality of swapped objects.

Proposition 8 (Two dual rho calculi in one universe; sketch)**.** *With the black reflection* $\overline{@} := \overline{\kappa} \circ \mathsf{s}$*, coherence* (3) *gives* $\overline{@} \cong \mathsf{s} \circ @ \circ \mathsf{s}$*: the two colours' quotes are swap-conjugate, one geometric reflection determining the other. The knotted topos carries two dual rho calculi at once, recovering the player/opponent duality of [1, Cor. 2] at the level of topoi — the interaction structure the game-semantic full abstraction of [10, 11] required.*

3.3 The behaviour endofunctor and the process object

Definition 9 (Internal behaviour endofunctor and process object)**.** Let $\partial T_K \in \mathcal{K}$ be the object of Fire context-labels (§4.4 identifies it with the site). Define $\widehat{\mathsf{B}}\colon \mathcal{K} \to \mathcal{K}$, $\widehat{\mathsf{B}}(R) = \mathcal{P}(\partial T_K \times R)$, with $\mathcal{P}$ the power object; its commitment refinement $\widehat{\mathsf{B}}^\flat$ — finite powerobject, products, coproducts, the name-abstraction functor $[\mathcal{N}](-)$ — presents the same behaviour finitely, after Milner [15, 2]. The *process object* is $\mathsf{Proc} := \nu\widehat{\mathsf{B}}$, the final $\widehat{\mathsf{B}}$-coalgebra internal to $\mathcal{K}$, with $\mathsf{Name} := \mathsf{s}(\mathsf{Proc})$ presented as $\mathcal{A}$ via κ. For $\widehat{\mathsf{B}}^\flat$ the final coalgebra exists at set size on the finitary core; for the full $\widehat{\mathsf{B}}$ it is the class-sized, colour-blind non-well-founded object of Remark 6, whose alternating-colour descent supplies the recursion of process behaviour for free, where Aczel's non-well-founded model imports it by axiom [28].

Proposition 10 (Finality internalised; sketch)**.** *For the weak-pullback-preserving $\widehat{\mathsf{B}}^\flat$, internal coalgebraic bisimilarity is the kernel of the unique morphism to Proc; hence on Proc itself two elements are context-bisimilar iff internally equal. Bisimulation is the equality of the process object in $\mathcal{K}$.*

Proof sketch. $\widehat{\mathsf{B}}^\flat$ is assembled from constants, products, coproducts, finite powerobject and the abstraction functor, each preserving weak pullbacks internally; universal coalgebra [17] identifies behavioural equivalence with the kernel of $!\colon C \to \nu\widehat{\mathsf{B}}^\flat$. This is [2, Lem. 17, Cor. 20] read internally to $\mathcal{K}$; the internal weak-pullback preservation of the abstraction functor and the agreement of $\nu\widehat{\mathsf{B}}^\flat$ with the class-sized $\nu\widehat{\mathsf{B}}$ on equality are part of Obligation 4. □

3.4 The lift, as a theorem

Theorem 11 (The lift)**.** *The four-sorted model of [2] lifts to $\mathfrak{Top}$: there is a knotted topos $\mathcal{K}$, fibred over the four-sort index category $\mathcal{G}$, carrying geometric knots $\kappa, \overline{\kappa}$ and an involutive colour-swap s with $\mathsf{s} \circ \mathsf{s} \cong \mathrm{id}$, hence derived reflection $@ = \kappa \circ \mathsf{s}$ and $* = \mathsf{s} \circ \kappa^{-1}$ with $*@ \cong \mathrm{id}$ (Proposition 4); presented as the (bi)fixpoint of $\widehat{\mathsf{H}}$ (Construction 5), set-sized on the finitary core and class-sized in general (Remark 6); equipped with an internal behaviour endofunctor $\widehat{\mathsf{B}}$ whose final coalgebra Proc is an object of $\mathcal{K}$ in which context bisimulation is internal equality (Proposition 10). Modulo Obligations 1–4, $\mathcal{K}$ is the topos-level knotted universe.*

4 Application: from operational to denotational semantics

We now use Theorem 11 to transport the operational view of the finitely presentable GSLTs to a denotational one, fully abstractly.

4.1 MeTTaIL presents the finitely presentable GSLTs

Definition 12 (MeTTaIL as a presentation)**.** A MeTTaIL program [6] is a triple of a finite collection of syntactic types (a grammar), a finite collection of equations on types, and a finite collection of rewrites on types. This is exactly the data of a finitely presentable GSLT. Sending a program to the GSLT it specifies, and a MeTTaIL language map (a translation of types respecting equations and rewrites) to the induced GSLT morphism, gives a functor $\mathbf{MeTTaIL} \to \mathbf{GSLT}^{\mathrm{fp}}_{\sim}$ that is essentially surjective: *MeTTaIL presents the category of finitely presentable GSLTs.* We work with GSLTs through this presentation, writing rules in MeTTaIL and compiling them to core rho.

4.2 Desugaring MeTTaIL into rho

The schema for a base rewrite is (2)'s companion,

$$[\![L \Rightarrow R]\!](c) \;=\; \mathsf{for}([\![L]\!] \Leftarrow c)\{\, c\,!\,([\![R]\!])\,\}, \tag{5}$$

a receiver consuming a match of $[\![L]\!]$ at channel c and re-emitting $[\![R]\!]$ there. The MeTTaIL pattern-receive $\mathsf{for}([\![L]\!] \Leftarrow c)$ is the sugar; under the embedding into core rho it unfolds to nested single-name receives with name-equality guards [3, §2.2], and the equation component of the source program compiles to Church-encoded normalisation, i.e. to structural congruence

— colour-respecting, cost-free, *iso* in the target rather than motion. The channel c names the runtime location.

Definition 13 (Location channels)**.** A *location* is a finite path $\ell = \langle (f_1, i_1), \dots, (f_m, i_m) \rangle$ from the root of a term; step (f, i) descends into the i-th argument of an f-node, ε is the root, and $\ell \cdot (f, i)$ the child. The channel is the reflection $c(\ell) := \ulcorner \ell \urcorner \in A$; distinct locations give distinct channels by injectivity of $\ulcorner \cdot \urcorner$.

Remark 14 (Freshness by quoting). Distinct injected terms must not share a root, lest $a!(0) \mid a!(0)$ collapse to $a!(0)$ — the identification Proc's parallel is built to avoid [2, §5]. Freshness is supplied as rho supplies all freshness, by quoting [2, §2]: an injection publishes on a root $\ulcorner \rho \urcorner$ for a quoted nonce ρ, and locations within it are $\ulcorner (\rho, \ell) \urcorner$. No ν, no central allocator. We suppress the root below.

Construction 15 (The desugaring functor)**.** For a finitely presentable GSLT G presented in MeTTaIL, $[\![G]\!] = |_{(L \Rightarrow R) \in \mathcal{R}} [\![L \Rightarrow R]\!]$ is the parallel composition of the persistent installations of (5) at all matching locations (Appendix A), with equations compiled to structural congruence. A $\mathbf{GSLT}^{\mathrm{fp}}_{\sim}$-morphism goes to the induced rho-context map on installed locations.

Proposition 16 (Operational correspondence; sketch)**.** *The context-labelled transition system of $[\![t]\!]$ in $[\![G]\!]$ is bisimilar to the rewrite transition system of t in G: each base-rewrite firing of t at location ℓ is matched by a $\widehat{\mathsf{B}}$-transition of $[\![t]\!]$ labelled $c(\ell)$, and conversely. Hence $[\![-]\!]$ is a functor $\mathbf{GSLT}^{\mathrm{fp}}_{\sim} \to \widehat{\mathsf{B}}\text{-}\mathbf{Coalg}(\mathcal{K})$ factoring through the bisimulation quotient.*

Proof sketch. $\Leftarrow$ is (5): a fire at ℓ is a rendezvous on $c(\ell)$. $\Rightarrow$ uses injectivity and plugging-stability of $c(\cdot)$ (Definition 13), so no spurious rendezvous occurs and the minimal enabling contexts are exactly the location channels; the labels coincide with ∂T_K, and the context-bisimulation game [2, Def. 3] is won by the identity on locations. Full statement: Obligation 3. $\square$

Remark 17 (Non-optimality). This desugaring re-inspects symbols enclosing a redex, failing the symbol-once condition (O1) of [3] while satisfying (O2),(O3) trivially — it is the "reflect the context verbatim" choice of [3, §3.2] lifted to the full runtime location. The optimal set-automaton scheme [29] recovers (O1); its deferred channel-naming correction and the present scheme induce the *same* context-labelled transition system, so Proposition 16 and all below are indifferent to the choice.

4.3 Lifting the rho denotation, and full abstraction

By Proposition 16 a presented GSLT denotes a $\widehat{\mathsf{B}}$-coalgebra; by finality (Proposition 10) it denotes a sub-coalgebra of $\mathsf{Proc} = \nu\widehat{\mathsf{B}}$ in $\mathcal{K}$. Write $[\![-]\!]_{\mathcal{K}}$ for the composite.

Theorem 18 (Fully abstract denotation)**.** *For every finitely presentable GSLT G and processes P, Q of G,*

$$[\![P]\!]_{\mathcal{K}} = [\![Q]\!]_{\mathcal{K}} \ \text{ in } \mathcal{K} \qquad \Longleftrightarrow \qquad P \sim Q,$$

where $\sim$ is context bisimulation. The denotation is compositional — a homomorphism for the GSLT's constructors — and $\sim$ is a congruence by construction (the context labels are idem-pushouts [13, 14, 2] so this is full abstraction with respect to a congruence, with no closure step.

Proof sketch. $[\![-]\!]_{\mathcal{K}}$ is the unique morphism to the final coalgebra (Proposition 10); its kernel is internal coalgebraic bisimilarity, which by [2, Lem. 16] read internally is context bisimilarity. Compositionality is the homomorphism property of the rho denotation [2, Thm. 15] transported along the desugaring (Proposition 16); congruence is [2, Prop. 4]. The whole is [2, Thm. 18] internalised to $\mathcal{K}$. $\square$

Corollary 19 (Milner answered, uniformly)**.** *Every finitely presentable classical model of computation receives a fully abstract denotational semantics in the knotted topos: the λ-calculus and hence PCF, CCS, the π-calculus, the ambient calculus, SKI, Turing machines, and finite term-rewriting systems, with universal algebra as the dynamics-free slice [4]. The model is the bisimulation quotient itself, so it is fully abstract by construction; the failure that defeats the Scott model of PCF [7, 8] — a model element with no realising program — cannot occur. The residual work per language is the identification of context bisimulation with that language's own observational equivalence (e.g. contextual equivalence for PCF): this is the barbed-congruence calibration of Obligation 5, and recasts the classical full-abstraction problem from a model-*construction *problem into a model-*calibration *one. The game-semantic models of [10, 11] are the equivariant, colour-respecting fragment (Remark 7), their player/opponent structure the red/black knot.*

4.4 The keystone: OSLF is the internal logic

Definition 20 (OSLF classifying topos and internal modalities)**.** For a presented GSLT G, let $\partial T_K(G)$ be its Fire context-labels, a site under the minimal-context (idem-pushout) covers; put $\mathrm{Cl}(G) := \mathrm{Sh}(\partial T_K(G))$. For each label F let $\langle F\rangle\colon \Omega \to \Omega$ be the internal operator on the subobject classifier sending φ to "some F-labelled transition reaches a state satisfying φ", defined by image along the F-component of the coalgebra structure map. The family $\{\langle F\rangle\}_{F\in\partial T_K}$ is the internal Hennessy–Milner logic of G. The assignment Cl extends to a 2-functor $\mathbf{GSLT}_\sim \to \mathfrak{Top}$, sending a bisimulation-preserving translation to a geometric morphism: the *topos-theoretic presentation of the semantic category of GSLTs*, with Cl(rho) $= \mathcal{K}$ the universal target through which every desugaring factors.

Theorem 21 (Adequacy is classification; sketch)**.** *The internal equality of* Proc *coincides with agreement under all internal modal formulae:*

$$[\![P]\!]_{\mathcal{K}} = [\![Q]\!]_{\mathcal{K}} \iff P \sim Q \iff \forall\varphi.\ (P \models \varphi \Leftrightarrow Q \models \varphi).$$

Equivalently the subobject classifier Ω of $\mathcal{K}$, with the modalities of Definition 20, classifies bisimilarity*: the OSLF-generated Hennessy–Milner logic is the internal logic of the knotted topos. The three readings of ∂T_K become site (Placing), labels (Firing), and modal indices (Witnessing) — three roles of one object of the topos. "Topos-theoretic presentation of GSLTs" and "OSLF adequacy" are one statement.*

Proof sketch. The first equivalence is Theorem 18; the second is OSLF adequacy [2, Cor. 19] read internally. That the internal modal theory is the subobject-classifier logic of $\mathrm{Sh}(\partial T_K)$ is Obligation 4. □

4.5 The risk ledger, lifted

A finitely presentable GSLT with finite synchronisation trees is a hereditarily finite object of the finitary-core topos, where $\nu\widehat{\mathsf{B}}^\flat$ is literal and set-sized; its commitment is the black empty set and one successor, no infinite risk. Recursion — reached through reflection, with no primitive replication — is a non-well-founded inhabitant of Proc, available only in the infinitary reading, which by Remark 6 costs exactly one ω, black infinity, and lives in the class-sized framework. No primitive name is ever reintroduced. "Finitely presentable" is therefore not a blemish on Corollary 19 but the exact regime in which the accounting is cleanest: the finite presentation is the finitary core, and only genuine recursion crosses into the one-ω ledger.

5 Toward a rigorous construction

In the manner of [1, §8] and [2, §8]. For the finitary core each obligation reduces to a statement about hereditarily finite data.

Obligation 1 (Existence of the knotted topos)**.** Exhibit the (bi)fixpoint of $\widehat{\mathsf{H}}$ of (4) in $\mathfrak{Top}$: for the finitary $\mathcal{P}$ as the set-sized colimit/limit of the ω-chain, each $\widehat{\mathsf{H}}^n(\varnothing)$ a topos; for the full $\mathcal{P}$ in the class-sized framework. The natural route is a bicategorical algebraic set theory [19, 20, 22] run two-sorted on $\widehat{\mathsf{H}}$, mirroring [1, §8.1]; an explicit alternative is the *knotted site*, two sites each of whose atom-objects are the other's representables.

Obligation 2 (Coherence of swap and knots)**.** Verify the 2-level equational theory: s an involution and a homomorphism up to coherent isomorphism, the coherence 2-cell (3) satisfying the triangle and pentagon identities, $\kappa, \overline{\kappa}$ equivalences. Then $@, *$ are derived and Proposition 4 is a calculation; the source papers' collapse (knots literal identities) is a localisation under which the reflection law survives.

Obligation 3 (Operational correspondence of the desugaring)**.** Promote Proposition 16 to a theorem: define the desugaring of Appendix A precisely, including persistence and the Church-encoded equation component, and prove bisimilarity of the two transition systems with labels matched by location channels; confirm independence from the channel intension (Remark 17).

Obligation 4 (The classifier classifies bisimilarity)**.** Prove Theorem 21: that the internal modal theory of Definition 20 coincides with the subobject-classifier logic of $\mathrm{Sh}(\partial T_K)$, that $\nu\widehat{\mathsf{B}}^\flat$ agrees with the class-sized $\nu\widehat{\mathsf{B}}$ on equality, and hence that Ω with its modalities classifies bisimilarity — internal OSLF adequacy, the keystone of Theorem 21.

Obligation 5 (Barbed-congruence calibration)**.** For each presented GSLT, calibrate context bisimulation $\sim$ against the language's own observational equivalence: reduction-barbed congruence for the process calculi, contextual equivalence at ground type for PCF, $\beta\eta$ or applicative bisimilarity for the λ-calculus. This is the residual of Corollary 19 — the classical full-abstraction question recast as calibration — and is expected tight where names are quoted structure and there is no ν [2, Obl. 6].

Obligation 6 (Functoriality on the encoded regime)**.** Prove Cl functorial on the *encoded* regime of $\mathbf{GSLT}_\sim$-morphisms — a source K implemented by target K's plus structural context (π-into-rho, λ-into-SKI) — so that operational-correspondence encodings induce geometric morphisms [4].

Obligation 7 (Size calibration)**.** State the finitary/infinitary calibration: the finitary-core $\mathcal{K}$ set-sized and interpretable in hereditarily finite data (equiconsistent with arithmetic [1, §8.2]); the full $\mathcal{K}$ raising the target by one ω. The well-founded vs non-well-founded knob is the initial- vs final-fixpoint choice of Remark 6.

Obligation 8 (The ultrafilter metric, internally)**.** Internalise the ultrafilter metric on bisimulation classes [4]: construct the points of the internal modal algebra of $\mathrm{Cl}(G)$, equip them with the metric induced by the site's spatial reach, and recover the real/complex-weight dichotomy as classical/Born-rule structure. This is the door for the physics reading.

Obligation 9 (Placing the construction)**.** Locate the knotted-topos presentation among its neighbours: classifying topoi of geometric theories and the topos-as-bridge methodology [23]; the game-semantic models of PCF [10, 11, 12], of which the equivariant fragment is an instance; the presheaf and nominal coalgebraic models of name-passing (Fiore–Turi [24], Stark [25]), of which this is the reflective, ν-free instance with $\mathsf{Name} \simeq \mathsf{Proc}$; the bialgebraic semantics of Turi–Plotkin [18]; and the nominal/Schanuel topos [26]. The feature absent from all of these — $\mathsf{Name} \simeq \mathsf{Proc}$ via s, the second knot — is what the lift adds.

6 Conclusion

The principal result is the lift: the knotted set-theoretic universe rises one categorical level to a knotted topos, four sort-topoi tied by two geometric knots and an involutive colour-swap, the reflection law become the swap's involutivity, the behaviour functor and its final coalgebra recapitulated internally where colour-blind non-well-foundedness gives recursion and finality makes bisimilar processes equal. The application transports the operational semantics of the finitely presentable GSLTs to a denotational one: MeTTaIL presents that category, desugars to rho, and the rho denotation lifted into the knotted topos is fully abstract for context bisimulation. Because GSLTs subsume the classical models of computation, this is a fully abstract denotational semantics for all of them at once. The model is the bisimulation quotient itself, so full abstraction is by construction; what is proved is compositionality, and the context-labelled presentation makes bisimulation a congruence with no closure step. The keystone is that the OSLF-generated Hennessy–Milner logic is the internal logic of the knotted topos, adequacy the statement that the subobject classifier classifies bisimilarity, and the three readings of ∂T_K — Firing, Witnessing, Placing — its labels, modalities and site. Milner's lament was that a denotational model can be too coarse or too fine for its language; the knotted topos answers that the right denotational object is neither chosen nor guessed but constructed as the interaction quotient, and that one such construction serves every finitely presentable model of computation. The risk ledger is unchanged: finite presentations cost nothing beyond the empty set, recursion costs one ω, and not one primitive name is ever spent.

A The desugaring, in clauses

We record $[\![\cdot]\!]$ used in §4.2, threading the location ℓ (Definition 13); $c(\ell) = \ulcorner \ell \urcorner$ is its channel and $\ell \cdot (f, i)$ the child location. The target is core rho; MeTTaIL pattern-receives abbreviate nested single-name receives with name-equality guards [3, §2.2], and equations are Church-encoded normalisations suppressed here.

Terms. For a constructor f of arity n,

$$[\![f(t_1, \dots, t_n)]\!]_\ell \;=\; c(\ell)\,!\,(\,\underline{f}\,) \;\Big|\; \Big|_{i=1}^{n} \;[\![t_i]\!]_{\ell\cdot(f,i)},$$

publishing the head tag $\underline{f}$ at the node's channel and installing each argument at its child location. A nullary constructor publishes only its tag; a schema variable installs nothing.

Base rewrites. For $L \Rightarrow R$ with $\mathrm{hd}(L) = f$,

$$[\![L \Rightarrow R]\!] \;= \Big|_{\ell\,:\,\mathrm{hd}=f} \;\mathsf{for}([\![L]\!] \Leftarrow c(\ell))\{\; c(\ell)\,!\,([\![R]\!]_\ell) \;\Big|\; [\![L \Rightarrow R]\!] \;\},$$

the inner copy re-installing the listener after each fire by the reflection idiom of [2, §2.1] (no replication). Bound names of $[\![L]\!]$ are the hole-fillers, delivered on $c(\ell)$.

Contextual rewrites. For $L_1 \Rightarrow R_1, \dots, L_n \Rightarrow R_n \Rightarrow K \Rightarrow K'$ with outer context K at ℓ,

$$\mathsf{let}\ c = c(\ell)\ \mathsf{in}\ \mathsf{for}(([\![L_1]\!], \dots, [\![L_n]\!]) \Leftarrow c)\{\; c\,!\,([\![K']\!]([\![R_1]\!], \dots, [\![R_n]\!])) \;\},$$

blocking until the n hole-fillers arrive, one per inner location $\ell \cdot (K, i)$, each a distinct channel by Definition 13; the send emits the rewritten outer right-hand side at ℓ.

A whole GSLT. $[\![G]\!] = |_{(L \Rightarrow R) \in \mathcal{R}} [\![L \Rightarrow R]\!]$, with equations compiled to structural congruence. A term t is run by injecting $[\![t]\!]_\varepsilon$ on a fresh root (Remark 14).

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